

Calculate the derivative:

$$1) f(x) = \frac{\sin x}{2 + \cos x} \quad u' = \cos x \quad u = \sin x \\ v' = -\sin x \quad v = 2 + \cos x$$

$$f'(x) = \frac{\cos x (2 + \cos x) - \sin x (-\sin x)}{(2 + \cos x)^2}$$

$$f'(x) = \frac{2\cos x + \cos^2 x + \sin^2 x}{(2 + \cos x)^2}$$

$$f'(x) = \frac{1 + 2\cos x}{(2 + \cos x)^2}$$

$f - v \cos$

